

Convex Optimization Problems

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Outline

- 1 Optimization Problems
- 2 Convex Optimization
- 3 Quasi-Convex Optimization
- 4 Classes of Convex Problems: LP, QP, SOCP, SDP

Optimization Problems in Standard Form I

$$\begin{array}{ll} \underset{\substack{x \in \mathbb{R}^n}}{\text{minimize}} & f_0(\mathbf{x}) \\ \text{subject to} & f_i(\mathbf{x}) \leq 0 \quad i = 1, \dots, m \\ & h_i(\mathbf{x}) = 0 \quad i = 1, \dots, p \end{array}$$

 $\mathbf{x} = (x_1, \dots, x_n)$ is the optimization variable

 $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective function

 $f_i : \mathbb{R}^n \rightarrow \mathbb{R} \quad i = 1, \dots, m$ are the inequality constraint functions

 $h_i : \mathbb{R}^n \rightarrow \mathbb{R} \quad i = 1, \dots, p$ are the equality constraint functions

Optimization Problems in Standard Form II

Feasibility:

- a point $x \in \text{dom } f_0$ is feasible if it satisfies all the constraints and infeasible otherwise
- a problem is feasible if it has at least one feasible point and infeasible otherwise.

Optimal value:

$$p^* = \inf \{ f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p \}$$

- $p^* = \infty$ if problem is infeasible (no x satisfies the constraints)
- $p^* = -\infty$ if problem is unbounded below

Optimal solution: x^* such that $f(x^*) = p^*$ (and x^* feasible).

optimal solution

optimal value

Global and Local Optimality

globally

$$X_{\text{opt}} = \{x \mid f_0(x) = p^*\}$$

- A feasible x is **optimal** if $f_0(x) = p^*$; X_{opt} is the set of optimal points.
- A feasible x is **locally optimal** if it is optimal within a ball, i.e., there is an $R > 0$ such that x is optimal for

$$\begin{array}{ll} \underset{z}{\text{minimize}} & f_0(z) \\ \text{subject to} & f_i(z) \leq 0 \quad i = 1, \dots, m \\ & h_i(z) = 0 \quad i = 1, \dots, p \\ & \|z - x\|_2 \leq R \end{array}$$

Example:

$$\lim_{x \rightarrow 0} \frac{1}{x} = \infty$$

- $f_0(x) = 1/x$, $\text{dom } f_0 = \mathbb{R}_{++}$: $p^* = 0$, no optimal point
- $f_0(x) = x^3 - 3x$: $p^* = -\infty$, local optimum at $x = 1$.

$$f_0'(x) = 0 \Rightarrow x^* = 1$$

Implicit Constraints

- The standard form optimization problem has an explicit constraint:

$$\mathbf{x} \in \mathcal{D} = \bigcap_{i=0}^m \underbrace{\text{dom } f_i} \cap \bigcap_{i=1}^p \underbrace{\text{dom } h_i}$$

- $\mathcal{D}$ is the domain of the problem

- The constraints $f_i(\mathbf{x}) \leq 0, h_i(\mathbf{x}) = 0$ are the explicit constraints

- A problem is unconstrained if it has no explicit constraints

- Example:

$$\underset{\mathbf{x} \in \mathbb{R}^n}{\text{minimize}} \quad \log(\underbrace{b - \mathbf{a}^T \mathbf{x}})$$

is an unconstrained problem with implicit constraint $b > \mathbf{a}^T \mathbf{x}$

$$b - \mathbf{a}^T \mathbf{x} > 0 \quad \mathbf{x} \geq 0$$

Feasibility Problem

- Sometimes, we don't really want to minimize any objective, just to find a feasible point:

$$\begin{array}{ll} \text{find} & \mathbf{x} \\ \text{subject to} & f_i(\mathbf{x}) \leq 0 \quad i = 1, \dots, m \\ & h_i(\mathbf{x}) = 0 \quad i = 1, \dots, p \end{array}$$

- This feasibility problem can be considered as a special case of a general problem:

$$\begin{array}{ll} \text{minimize} & 0 \\ \text{subject to} & f_i(\mathbf{x}) \leq 0 \quad i = 1, \dots, m \\ & h_i(\mathbf{x}) = 0 \quad i = 1, \dots, p \end{array}$$

where $p^* = 0$ if constraints are feasible and $p^* = \infty$ otherwise.

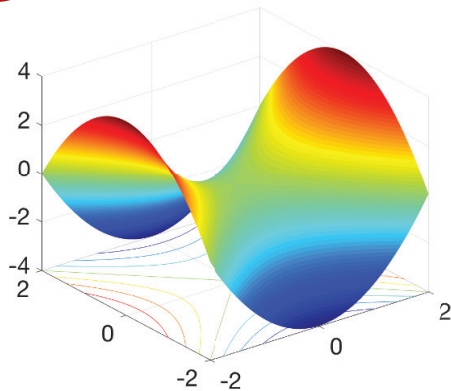
①

②

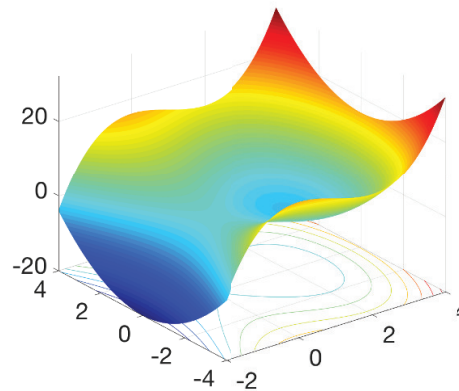
Stationary Points

Given a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, a point $x \in \mathbb{R}^n$ is called

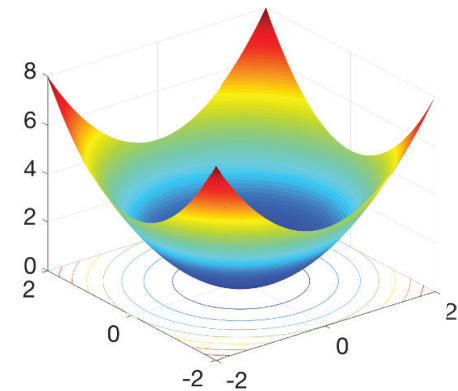
- A **stationary point**, if $\nabla f(x) = 0$;
- A **local minimum**, if x is a stationary point and there exists a neighborhood $\mathcal{B} \subseteq \mathbb{R}^n$ of x such that $f(x) \leq f(y)$ for any $y \in \mathcal{B}$;
- A **global minimum**, if x is a stationary point and $f(x) \leq f(y)$ for any $y \in \mathbb{R}^n$;
- **Saddle point**, if x is a stationary point and for any neighborhood $\mathcal{B} \subseteq \mathbb{R}^n$ of x , there exist $y, z \in \mathcal{B}$ such that $f(z) \leq f(x) \leq f(y)$ and $\lambda_{\min}(\nabla^2 f(x)) \leq 0$.



(a) strict saddle



(b) local minimum

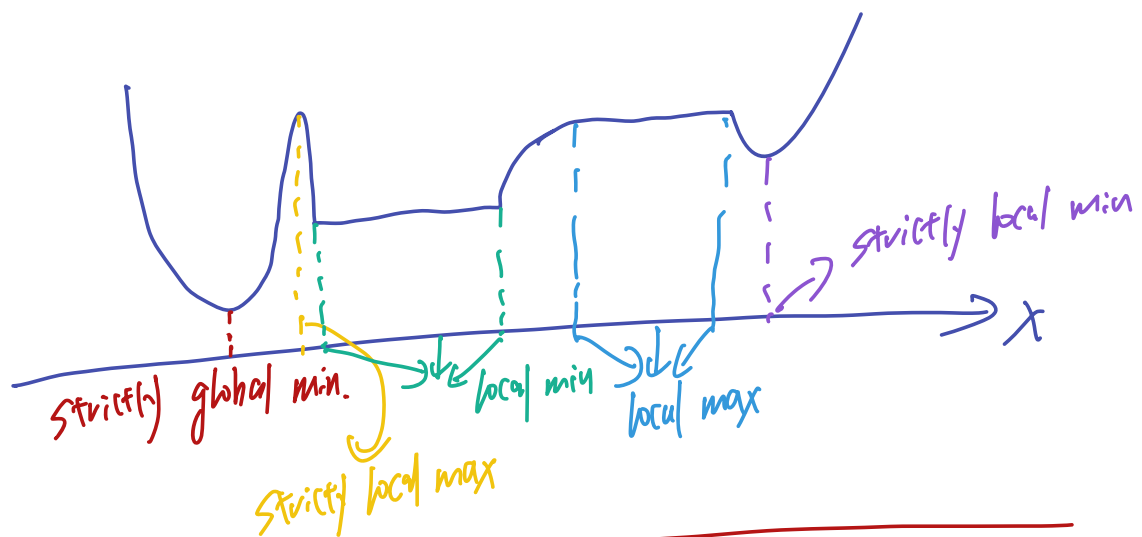


(c) global minimum

stationary points could be

{ local minima, local maxima
saddle points, global minima

$\lambda_{\min}(\nabla^2 f(x)) \begin{cases} > 0, & \text{local minimum} \\ \geq 0, & \text{local minimum/saddle points} \\ < 0, & \text{strict saddle point} \end{cases}$



minimize $f(x)$ — (1)
subject to $x \in \mathcal{U}$
 $\downarrow$ feasible set

A point $\bar{x}$ is said to be a

① local minimum, if $\bar{x} \in U$ and $\exists \epsilon > 0$, s.t.
 $f(\bar{x}) \leq f(x)$, $\forall x \in B(\bar{x}, \epsilon) \cap U$

② strictly local minimum, if $\bar{x} \in U$ and
if $\bar{x} \in U$, s.t.

$$\underline{f(\bar{x}) < f(x)}, \quad \forall x \in B(\bar{x}, \epsilon) \cap U$$

$$x \neq \bar{x}$$

③ global minimum if $\bar{x} \in U$ and if

$$f(\bar{x}) \leq f(x), \quad \forall x \in U$$

④ strictly global minimum, if $\bar{x} \in U$, and if

$$\underline{f(\bar{x}) < f(x)}, \quad \forall x \in U, \quad x \neq \bar{x}$$

First-order optimality conditions.

$$\underset{x}{\text{minimize}} \quad f(x)$$

Definition (descent direction): consider a function

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ and a point $x \in \mathbb{R}^n$. A direction $d \in \mathbb{R}^n$ is a descent direction at x , if $\exists \bar{\alpha} > 0$, s.t.

$$f(x + \alpha d) < f(x), \quad \forall \alpha \in (0, \bar{\alpha})$$

step size

search direction

Lemma 1: Consider a point $x \in \mathbb{R}^n$ and a continuously differentiable function f . Then, any direction d that satisfies

$$\langle \nabla f(x), d \rangle < 0$$

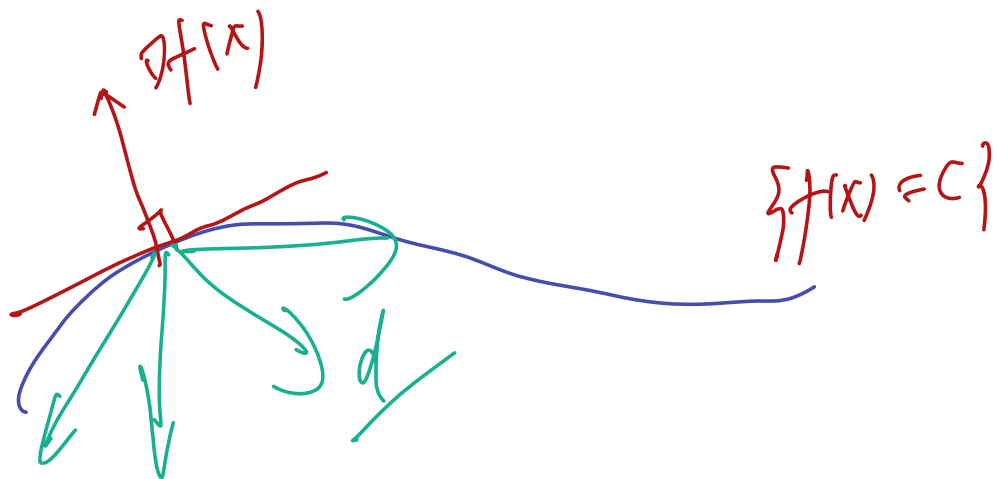
is a descent direction

(In particular, $-\nabla f(x)$ is a descent direction if it is non-zero)

$$X_{n+1} = X_n - \alpha_k \nabla f(X_n)$$

↓
stepsize (learning rate in ML)

Gradient Descent Method



proof: let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(\alpha) = f(\underbrace{X}_{\text{fixed}} + \alpha \underbrace{d}_{\text{fixed}})$$

$g'(\alpha) = d^T \nabla f(X + \alpha d)$
we can use Taylor expansion to write

$$g(t) = g(0) + g'(0)t + o(t)$$

$$\Leftrightarrow f(x+td) = f(x) + td^T \nabla f(x) + o(t)$$

$$\Leftrightarrow \frac{f(x+td) - f(x)}{t} = \underbrace{d^T \nabla f(x)}_{< 0} + \underbrace{\frac{o(t)}{t}}$$

Since $\lim_{t \downarrow 0} \frac{|o(t)|}{t} = 0$, there exists $\bar{t}$,
 s.t. $\forall t \in (0, \bar{t})$, we have $\frac{|o(t)|}{t} < \frac{1}{2} |d^T \nabla f(x)|$

Since $d^T \nabla f(x) < 0$ by assumption, we conclude

$$\forall t \in (0, \bar{t})$$

$$\frac{f(x+td) - f(x)}{t} < \underbrace{\frac{1}{2} d^T \nabla f(x)}_{< 0} < 0$$

Remark: The converse of this lemma is not true.

e.g., $f(x) = x_1^2 - x_2^2$, $d = (0, 1)^T$, $\bar{x} = (1, 0)^T$

for $t \in \mathbb{R}$, we have

$$f(\bar{x} + td) - f(\bar{x}) = 1^2 - (0 + t^2) - 1^2 + 0 = \underline{-t^2} < 0$$

$\therefore$ based on definition, d is descent direction. but

$$\langle \nabla f(\bar{x}), d \rangle = (2, 0) \cdot (0, 1)^T = 0$$

First-order necessary condition for optimality.

Theorem: If $\bar{x}$ is an unconstrained local minimum of a differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, then

$$\boxed{\nabla f(\bar{x}) = 0}$$

Proof: if $\nabla f(\bar{x}) \neq 0$, then $\exists i$, s.t. $\frac{\partial f(x)}{\partial x_i} \neq 0$

Then, from lemma 9, either e_i or $-e_i$ is a descent direction.

(Here, e_i is the i -th standard basis vector)

$$e_i = (0, \dots, 0, \underbrace{1}_{i\text{-th}}, 0, \dots, 0)$$

$$\frac{\partial f(x)}{\partial x_i} \begin{cases} > 0, & d = -e_i \Rightarrow \langle \nabla f(x), d \rangle = -\frac{\partial f(x)}{\partial x_i} < 0 \\ < 0, & d = e_i \Rightarrow \langle \nabla f(x), d \rangle = \frac{\partial f(x)}{\partial x_i} < 0 \end{cases}$$

Therefore, $\bar{x}$ cannot be a local minimum.

Second-order necessary condition for local optimality

Theorem: If x^* is an unconstrained local minimizer of a twice differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$.

Then, in addition to $\nabla f(x^*) = 0$, we must have

$$\underline{\lambda_{\min}(\nabla^2 f(x^*)) \geq 0}, \quad \underline{\nabla^2 f(x^*)} \text{ (positive semidefinite)}$$

Proof: Consider some $y \in \mathbb{R}^n$. For $\delta > 0$, the second-order Taylor expansion of f around x^* gives

$$f(x^* + \delta y) = f(x^*) + \underbrace{\delta y^T \nabla f(x^*)}_{=0} + \frac{\delta^2}{2} y^T \nabla^2 f(x^*) y + o(\delta^2)$$

$$\underbrace{\frac{f(x^* + \delta y) - f(x^*)}{\delta^2}}_{\geq 0} = \frac{1}{2} y^T \nabla^2 f(x^*) y + \frac{o(\delta^2)}{\delta^2}$$

By definition of local optimality of x^* , the left hand side is non-negative for δ sufficiently small

$$\lim_{\delta \downarrow 0} \frac{1}{2} y^T \nabla^2 f(x^*) y + \frac{o(\delta^2)}{\delta^2} \geq 0$$

But, $\lim_{\delta \downarrow 0} \frac{o(\delta^2)}{\delta^2} = 0 \Rightarrow y^T \nabla^2 f(x^*) y \geq 0$
 $\Downarrow$
 $\nabla^2 f(x^*) \succeq 0$ can be arbitrary

remark: The converse of this theorem is not true

$$\nabla f(x^*) = 0, \quad \lambda_{\min}(\nabla^2 f(x^*)) = 0$$

$\Downarrow$
local minimum / saddle point

Second-order sufficient condition for local optimality

Theorem: Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable and there exists a point x^* such that

$$\nabla f(x^*) = 0, \quad \lambda_{\min}(\nabla^2 f(x^*)) > 0$$

then, x^* is strictly local minimum of f

proof: let $\lambda > 0$ be the minimum eigenvalue of

$\nabla^2 f(x^*)$. This implies that

$$\nabla^2 f(x^*) - \lambda I \succeq 0 \text{ PSD}$$

$$\frac{\lambda_1 - \lambda}{> 0}, \quad \dots, \quad \frac{\lambda_n - \lambda}{= 0}$$

$$y^T (\nabla^2 f(x^*) - \lambda I) y \geq 0, \quad \forall y \in \mathbb{R}^n$$

$\Downarrow$

$$y^T \nabla^2 f(x^*) y \geq \lambda \|y\|_2^2$$

once again, Taylor expansion yields

$$f(x^* + y) - f(x^*) = \underbrace{y^T \nabla f(x^*)}_{=0} +$$

$$\underbrace{\frac{1}{2} y^T \nabla^2 f(x^*) y}_{\geq \frac{\lambda}{2} \|y\|^2} + o(\|y\|^2)$$

$$\geq \frac{\lambda}{2} \|y\|^2 + o(\|y\|^2)$$

$$= \|y\|^2 \left(\frac{\lambda}{2} + \underbrace{\frac{o(\|y\|^2)}{\|y\|^2}} \right) \quad (y \neq 0)$$

since $\lim_{\|y\| \rightarrow 0} \frac{o(\|y\|^2)}{\|y\|^2} = 0$, $\exists \delta > 0$, s.t.

$$\left| \frac{o(\|y\|^2)}{\|y\|^2} \right| < \frac{\lambda}{2}, \quad \forall y \text{ with } \|y\| < \delta$$

therefore, $f(x^* + y) > f(x^*)$, $\forall \|y\| < \delta$

x^* is a strictly local minimum

remark: the converse of this theorem is not true

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Convex Optimization Problem

- Convex optimization problem in standard form:

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0 \quad i = 1, \dots, m \\ & \mathbf{Ax} = \mathbf{b}\end{array}$$

where $f_0, f_1, \dots, f_m$ are convex and equality constraints are affine.

- **Local and global optima:** any locally optimal point of a convex problem is globally optimal

- Most problems are not convex when formulated
- Reformulating a problem in convex form is an art, there is no systematic way

Example

- The following problem is nonconvex (why not?):

$$\begin{array}{ll}\underset{x}{\text{minimize}} & x_1^2 + x_2^2 \\ \text{subject to} & x_1/(1 + x_2^2) \leq 0 \\ & \underline{(x_1 + x_2)^2 = 0}\end{array}$$

- The objective is convex.
- The equality constraint function is not affine; however, we can rewrite it as $x_1 = -x_2$ which is then a linear equality constraint.
- The inequality constraint function is not convex; however, we can rewrite it as $x_1 \leq 0$ which again is linear.
- We can rewrite it as

$$\begin{array}{ll}\underset{x}{\text{minimize}} & x_1^2 + x_2^2 \\ \text{subject to} & x_1 \leq 0 \\ & x_1 = -x_2\end{array}$$

Global and Local Optimality

Any locally optimal point of a convex problem is globally optimal.

Proof: Suppose x is locally optimal (around a ball of radius R) and y is optimal with $f_0(y) < f_0(x)$. We will show this cannot be.

Just take the segment from x to y : $z = \theta y + (1 - \theta)x$.

Obviously the objective function is strictly decreasing along the segment since $f_0(y) < f_0(x)$:

$$\theta f_0(y) + (1 - \theta)f_0(x) < f_0(x) \quad \theta \in (0, 1]$$

Using now the convexity of the function, we can write

$$f_0(\theta y + (1 - \theta)x) < f_0(x) \quad \theta \in (0, 1]$$

Finally, just choose θ sufficiently small such that the point z is in the ball of local optimality of x , arriving at a contradiction.

Convex optimization

minimize $f(x)$

subject to $x \in \mathcal{U}$

f : convex function, $\mathcal{U}$ is a convex set

Theorem: any local minimum is also a globally minimum

Proof: Let $\bar{x}$ be a local minimum $\Rightarrow$

$\bar{x} \in \mathcal{U}$, and $\exists \epsilon > 0$, s.t.

$f(\bar{x}) \leq f(x)$, $\forall x \in B(\bar{x}, \epsilon)$

suppose for the sake of contradiction that

$\exists z \in \mathcal{U}$ with

$$f(z) < f(\bar{x}) \quad \text{--- } \textcircled{D}$$

1) Because of convexity of $\mathcal{U}$, we have

$$\lambda \bar{x} + (1-\lambda)z \in \mathcal{U}, \quad \forall \lambda \in (0,1)$$

2) Because of convexity of f , we have

$$\begin{aligned} f(\lambda \bar{x} + (1-\lambda)z) &\stackrel{\text{def.}}{\leq} \lambda f(\bar{x}) + (1-\lambda)f(z) \\ &\leq \lambda f(\bar{x}) + (1-\lambda)f(\bar{x}) \\ &= f(\bar{x}) \end{aligned}$$

as $\lambda \rightarrow 1$, $\lambda \bar{x} + (1-\lambda)z \rightarrow B(\bar{x}, \epsilon)$

contradicts local optimality of $\bar{x}$.

Optimality Criterion for Differentiable f_0 I

Minimum Principle: A feasible point x is optimal if and only if

$$\nabla f_0(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \geq 0 \quad \text{for all feasible } \mathbf{y}$$

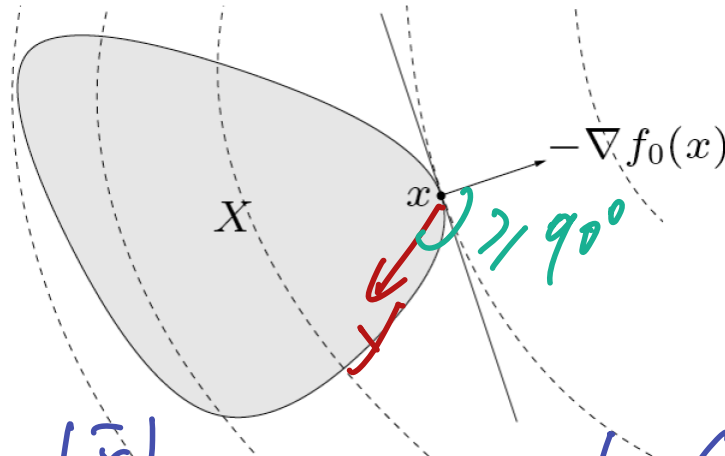
minimize $f_0(x)$
 $x \in \mathcal{R}^n$

$$\Rightarrow \partial f_0(\bar{x}) = 0$$

$$p_{\text{wert}} = y = \bar{x} - \sigma \cdot D_{f_0}(\bar{x})$$

$$\Rightarrow -\delta \| \nabla f_0(\bar{x}) \|^2 \geq 0$$

$$\Rightarrow Df_0(\bar{x}) = 0$$



↓ $\{x \mid f(x) = c\}$

$$\begin{array}{l} \text{minimize } f(x) \\ x \in \mathbb{R}^n \end{array}$$

Theorem: f is differentiable and convex. Any point $\bar{x}$ that satisfies $\nabla f(\bar{x}) = 0$, is a global minimum

Proof: from the first-order characterization of convexity, we have

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y$$

in particular,

$$f(y) \geq f(\bar{x}) + \underbrace{\nabla f(\bar{x})^T}_{=0} (y - \bar{x}), \quad \forall y$$

$\Downarrow$

$$f(y) \geq f(\bar{x}), \quad \forall y.$$

optimality conditions for convex optimization

$$\text{minimize } f(x)$$

$$\text{subject to } x \in \mathcal{C}$$

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, $\mathcal{C}$ is convex set.
Then a point is globally optimal, iff $x \in \mathcal{C}$, and

$$\langle \nabla f(x), y - x \rangle \geq 0, \quad y \in \mathcal{C}$$

Proof:

(Sufficiency) suppose $x \in \mathcal{C}$ satisfies

$$\langle \nabla f(x), y - x \rangle \geq 0, \quad \forall y \in \mathcal{C} \quad - (1)$$

By the first-order characterization of convexity, we have

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall y \in \mathcal{C} \quad - (2)$$

$$(1) + (2) \Rightarrow f(y) \geq f(x), \quad \forall y \in \mathcal{C}.$$

$$\Rightarrow x \text{ is globally optimal}$$

(necessary) suppose x is globally optimal,
but for some $y \in \mathcal{N}$, we have [descent direction]

$$\langle Df(x), y - x \rangle < 0 \quad \textcircled{x}$$

consider $g(t) = f(\underline{x + t(y - x)})$

$\mathcal{N}$ is convex $\Rightarrow \forall t \in [0, 1], x + t(y - x) \in \mathcal{N}$

observe that $g'(t) = (y - x)^T \nabla f(x + t(y - x))$

$$\Downarrow t = 0$$

$$g'(0) = \underline{(y - x)^T \nabla f(x)} < 0$$

$$\Downarrow$$

$$\exists \delta > 0, \text{ s.t. }, g(t) < g(0), \quad \forall t \in [0, \delta]$$

$$\Downarrow$$

$$f(x + t(y - x)) < f(x), \quad \forall t \in [0, \delta]$$

Optimality Criterion for Differentiable f_0 II

• **Unconstrained problem:** x is optimal iff

$$x \in \text{dom } f_0, \quad \nabla f_0(x) = 0$$

• **Equality constrained problem:** $\min_x f_0(x) \quad \text{s.t. } Ax = b$ (Lect. 4)
 x is optimal iff KK?

$$x \in \text{dom } f_0, \quad Ax = b, \quad \nabla f_0(x) + A^T \nu = 0$$

• **Minimization over nonnegative orthant:** $\min_x f_0(x) \quad \text{s.t. } x \succeq 0$
 x is optimal iff

$$x \in \text{dom } f_0, \quad x \succeq 0, \quad \begin{cases} \nabla_i f_0(x) \geq 0 & x_i = 0 \\ \nabla_i f_0(x) = 0 & x_i > 0 \end{cases}$$

$$\sum_i \lambda_i \nabla_i f_0(x) = \nabla f_0(x)^T x = 0$$

$$\text{minimize } f(x) \quad \text{subject to } x \geq 0$$

optimality condition

$$\langle \nabla f(x), y - x \rangle \geq 0, \quad \underline{x \geq 0, y \geq 0}$$

$\Downarrow$

$$\underline{\nabla f(x)^T y - \nabla f(x)^T x \geq 0}$$

$$\textcircled{1} \quad \nabla f(x)^T y = \sum_i \nabla f_i(x)^T y_i \quad (y_i \geq 0)$$

is unbounded below, unless $\nabla f(x) \geq 0$
 $(-\infty)$

$$\forall \nabla f_i(x) \geq 0$$

$\Downarrow$

$$- \nabla f(x)^T x \geq 0$$

$$\Downarrow \underline{\nabla f(x) \geq 0, x \geq 0}$$

$$\nabla f(x)^T x = 0$$

Equivalent Reformulations I

• Eliminating/introducing equality constraints:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0 \quad i = 1, \dots, m \end{array}$$

$$Ax = b$$

is equivalent to

$$\begin{array}{ll} \underset{z}{\text{minimize}} & f_0(Fz + x_0) \\ \text{subject to} & f_i(Fz + x_0) \leq 0 \quad i = 1, \dots, m \end{array}$$

where F and x_0 are such that $Ax = b \iff x = Fz + x_0$ for some z .

Equivalent Reformulations II

• Introducing slack variables for linear inequalities:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & \underline{a_i^T x \leq b_i} \quad i = 1, \dots, m \end{array}$$

is equivalent to

$$\begin{array}{ll} \underset{x, s}{\text{minimize}} & f_0(x) \\ \text{subject to} & \left\{ \begin{array}{l} a_i^T x + s_i = b_i \quad i = 1, \dots, m \\ s_i \geq 0 \end{array} \right. \end{array}$$

Handwritten notes in red:

- $\min$
- s.t.
- $C^T x$
- $Ax = b$
- $x \in C_1 \times C_2 \times C_3 \times \dots \times C_n$

Equivalent Reformulations III

• **Epigraph form:** a standard form convex problem is equivalent to

$$\begin{array}{ll}
 \underset{x, t}{\text{minimize}} & t \\
 \text{subject to} & f_0(x) - t \leq 0 \\
 & \underbrace{f_i(x) \leq 0}_{\text{convex set}} \quad i = 1, \dots, m \\
 & Ax = b
 \end{array}$$

$$\begin{array}{ll}
 \text{minimize} & f_0(x) \\
 \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\
 & Ax = b
 \end{array}$$

Equivalent Reformulations IV

• Minimizing over some variables:

minimize
 x, y

$f_0(x, y)$

jointly convex in (x, y)

subject to

$f_i(x) \leq 0 \quad i = 1, \dots, m$

is equivalent to

minimize
 x

$\tilde{f}_0(x)$

subject to

$f_i(x) \leq 0 \quad i = 1, \dots, m$

where $\tilde{f}_0(x) = \inf_y f_0(x, y)$

convex

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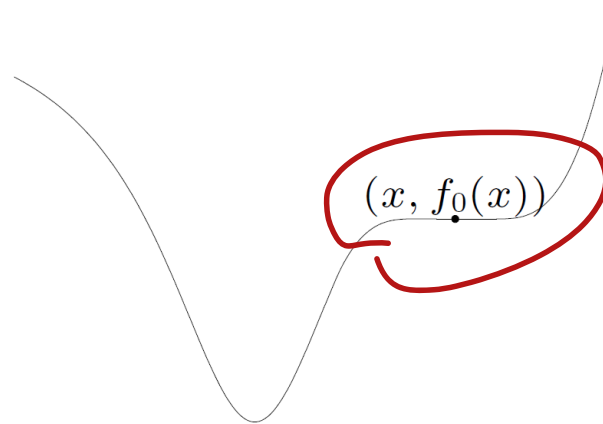
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Quasiconvex Optimization

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0 \quad i = 1, \dots, m \\ & Ax = b \end{array} \quad \left. \vphantom{\begin{array}{l} \text{minimize} \\ \text{subject to} \end{array}} \right\} \text{Convex}$$

where $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconvex and $f_1, \dots, f_m$ are convex

- 🦋 Observe that it can have locally optimal points that are not (globally) optimal:



Quasiconvex Optimization

- **Convex representation** of sublevel sets of a quasiconvex function f_0 : there exists a family of convex functions $\phi_t(\mathbf{x})$ for fixed t such that

$$f_0(\mathbf{x}) \leq t \iff \phi_t(\mathbf{x}) \leq 0$$

- **Example:**

minimize $\mathbf{x}$ $f_0(\mathbf{x}) = \frac{p(\mathbf{x})}{q(\mathbf{x})} \Rightarrow$ *find $\mathbf{x}$ subject to $f_0(\mathbf{x}) \leq t$*

with ~~p convex~~, ~~q concave~~, and $p(\mathbf{x}) \geq 0$, $q(\mathbf{x}) > 0$ on $\text{dom } f_0$. We can choose:

$$\phi_t(\mathbf{x}) = p(\mathbf{x}) - tq(\mathbf{x})$$

convex

- for $t \geq 0$, $\phi_t(\mathbf{x})$ is convex in $\mathbf{x}$
- $p(\mathbf{x})/q(\mathbf{x}) \leq t$ if and only if $\phi_t(\mathbf{x}) \leq 0$

Quasiconvex Optimization

Solving a quasiconvex problem via convex feasibility problems: the idea is to solve the epigraph form of the problem with a sandwich technique in t :

- for fixed t the epigraph form of the original problem reduces to a feasibility convex problem

convex feasibility / find x s.t.

$$\phi_t(x) \leq 0, \quad f_i(x) \leq 0 \forall i, \quad Ax \leq b$$

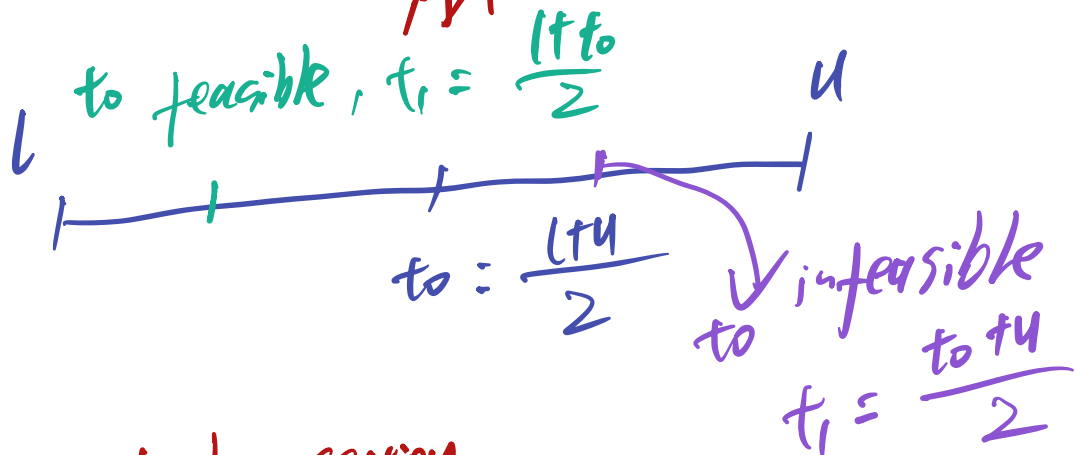
- if t is too small, the feasibility problem will be infeasible
- if t is too large, the feasibility problem will be feasible
- start with upper and lower bounds on t (termed u and l , resp.) and use a sandwich technique (bisection method): at each iteration use $t = (l + u)/2$ and update the bounds according to the feasibility or infeasibility of the problem.

$$\begin{array}{ll}
 \text{minimize} & f_0(x) \rightarrow \text{quasi-convex} \\
 \text{subject to} & f_i(x) \leq 0 \\
 & Ax = b
 \end{array}$$

$\Downarrow$ fixed t

$$\begin{array}{ll}
 \text{minimize} & \underline{t} \quad \text{constant} \\
 \text{subject to} & \underline{f_0(x) \leq t} \Rightarrow \underline{\phi_t(x) \leq 0} \\
 & \quad \quad \quad \text{convex} \\
 & f_i(x) \leq 0
 \end{array}$$

$$Ax = b$$



$$\# \text{ steps of bi-section: } \log_2 \left\lceil \frac{U-L}{\epsilon} \right\rceil$$

Outline

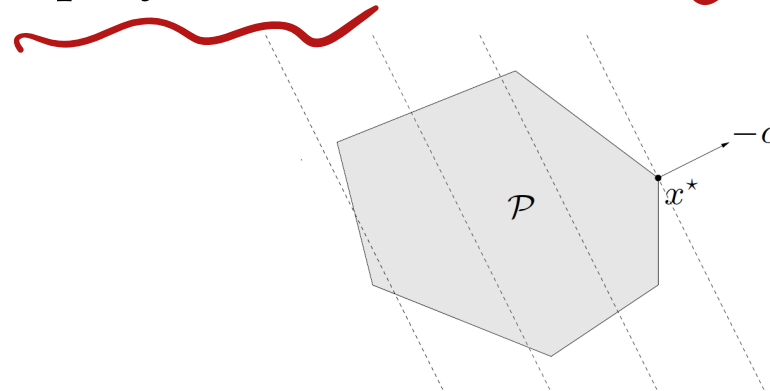
- 1 Optimization Problems
- 2 Convex Optimization
- 3 Quasi-Convex Optimization
- 4 Classes of Convex Problems: LP, QP, SOCP, SDP

Linear Programming (LP)

$$\begin{array}{ll}\underset{x}{\text{minimize}} & c^T x + d \\ \text{subject to} & \left. \begin{array}{l} Gx \leq h \\ Ax = b \end{array} \right\}\end{array}$$

• Convex problem: affine objective and constraint functions.

• Feasible set is a polyhedron:



ℓ_1 - and ℓ_∞ - Norm Problems as LPs I

• ℓ_∞ -norm minimization:

minimize
 x

subject to

$$\|x\|_\infty$$

$$Gx \leq h$$

$$Ax = b$$

$$:= \max_i |x_i|$$

↓ epi graph

is equivalent to the LP

minimize
 t, x

subject to

$$-t1 \preceq x \preceq t1$$

$$Gx \leq h$$

$$Ax = b$$

minimize t
 x, t

subject to $\|x\|_\infty \leq t$

$$Gx \leq h$$

$$Ax = b$$

$$\|x\|_\infty \leq t$$

$$\Downarrow$$

$$\max_i |x_i| \leq t$$

$$\Downarrow$$

$$|x_i| \leq t, \forall i=1, \dots, n$$

ℓ_1 - and ℓ_∞ - Norm Problems as LPs II

ℓ_1 -norm minimization:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \|x\|_1 \\ \text{subject to} & Gx \leq h \\ & Ax = b \end{array}$$

is equivalent to the LP

$$\begin{array}{ll} \underset{t, x}{\text{minimize}} & \sum_i t_i \\ \text{subject to} & -t \preceq x \preceq t \\ & Gx \leq h \\ & Ax = b \end{array}$$

$x \in C$
 $x \in C$
 $s.t. \quad |x_i| = t_i, i=1, \dots, n$
 (exercise)

Examples: Chebyshev Center of a Polyhedron I

- Chebyshev center of a polyhedron

$$\mathcal{P} = \{x \mid \underline{a_i^T x \leq b_i, i = 1, \dots, m}$$

is center of largest inscribed ball

$$\mathcal{B} = \{x_c + u \mid \|u\| \leq r\}$$

- Let's solve the problem

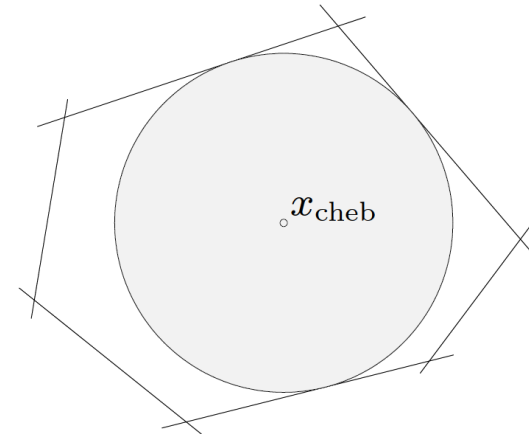
$$\text{maximize}_{r, x_c} \quad r$$

$$\text{subject to} \quad x \in \mathcal{P} \text{ for all } x = x_c + u \text{ with } \|u\| \leq r$$

- Observe that $a_i^T x \leq b_i$ for all $x \in \mathcal{B}$ if and only if

$$\sup_u \{a_i^T (x_c + u) \mid \|u\| \leq r\} \leq b_i$$

$$a_i^T x_c + \underline{a_i^T u} \leq a_i^T x_c + \|a_i\| \|u\| \leq a_i^T x_c + r \|a_i\|$$



Examples: Chebyshev Center of a Polyhedron II

- Using Schwartz inequality, the supremum condition can be rewritten as

$$\underline{a_i^T x_c + r \|a_i\|_2 \leq b_i} \quad x_c, r$$

- Hence, the Chebyshev center can be obtained by solving:

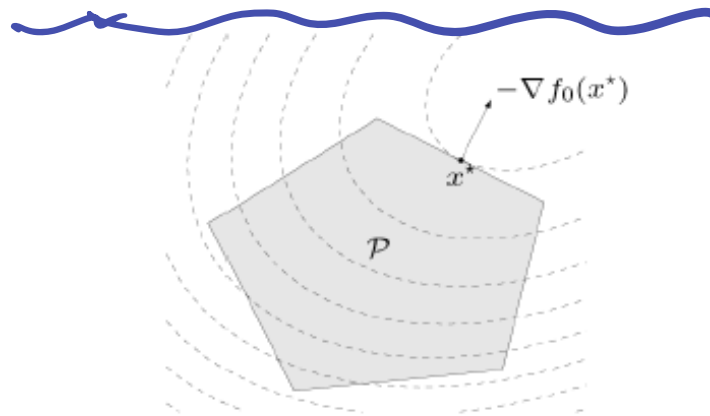
$$\begin{array}{ll} \text{maximize} & r \\ \text{subject to} & a_i^T x_c + r \|a_i\|_2 \leq b_i, \quad i = 1, \dots, m \end{array}$$

which is an LP.

Quadratic Programming (QP)

$$\begin{array}{ll}\underset{x}{\text{minimize}} & (1/2) \mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + r \\ \text{subject to} & \mathbf{G} \mathbf{x} \leq \mathbf{h} \\ & \mathbf{A} \mathbf{x} = \mathbf{b}\end{array}$$

- Convex problem (assuming $\mathbf{P} \in \mathbb{S}_+^n \succeq \mathbf{0}$): convex quadratic objective and affine constraint functions.
- Minimization of a convex quadratic function over a polyhedron:



Quadratically Constrained QP (QCQP)

$$\begin{aligned}
 &\underset{x}{\text{minimize}} && (1/2) x^T P_0 x + q_0^T x + r_0 \\
 &\text{subject to} && (1/2) x^T P_i x + q_i^T x + r_i \leq 0 \quad i = 1, \dots, m \\
 &&& Ax = b
 \end{aligned}$$

- Convex problem (assuming $P_i \in \mathbb{S}_+^n \succeq 0$): convex quadratic objective and constraint functions.

$$x^T P_i x = \text{trace}(P_i \underline{M}) \quad \text{convex in } M$$

$$M = XX^T, \quad \underline{\text{rank}}(M) = 1 \quad \text{non-convex}$$

Second-Order Cone Programming (SOCP)

$$\begin{array}{ll}
 \underset{x}{\text{minimize}} & f^T x \\
 \text{subject to} & \|A_i x + b_i\| \leq c_i^T x + d_i \quad i = 1, \dots, m \\
 & \underline{Fx = g}
 \end{array}$$

second-order cone

• Convex problem: linear objective and second-order cone constraints

~~• For A_i row vector, it reduces to an LP~~

~~• For $c_i = 0$, it reduces to a QCQP~~

• More general than QCQP and LP

$$\left\{ \begin{array}{l} z_i = A_i x + b_i \\ y_i = c_i^T x + d_i \end{array} \right\}$$

$$(z_i, y_i) \in \mathcal{K}$$

second-order cone

Robust LP as an SOCP

• Sometimes, the parameters of an optimization problem are imperfect

• Consider the robust LP:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & a_i^T x \leq b_i \quad \forall a_i \in \mathcal{E}_i, \quad i = 1, \dots, m \end{array}$$

where $\mathcal{E}_i = \{\bar{a}_i + P_i u \mid \|u\| \leq 1\}$

uncertainty set

• It can be rewritten as the SOCP:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \|P_i^T x\|_2 \leq b_i \quad i = 1, \dots, m \end{array}$$

$$\sup a_i^T x = \sup \{ (\bar{a}_i + P_i u)^T x \mid \|u\| \leq 1 \}$$

$$= \bar{a}_i^T x + \underbrace{u^T P_i^T x}_{\leq \|P_i^T x\|_2} \leq \bar{a}_i^T x + \|P_i^T x\|_2 \underbrace{\|u\|}_{\leq 1} =$$

Generalized Inequality Constraints

- Convex problem with generalized inequality constraints:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & \underbrace{f_i(x) \preceq_{K_i} \mathbf{0}}_{Ax = b} \quad i = 1, \dots, m \end{array}$$

where f_0 is convex and f_i are K_i -convex w.r.t. proper cone K_i

- It has the same properties as a standard convex problem

• **Conic form problem:** special case with affine objective and constraints:



$$\begin{array}{ll} \underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & \underbrace{Fx + g \preceq_K \mathbf{0}}_{Ax = b} \end{array}$$

conic optimization

① minimize $C^T X$, subject to $AX=b$, $X \in K$

tractable conic optimization (self-dual cone)

① The nonnegative orthant, $R_+^n \ni LP$

② The second-order cone, $Q^n = \{(X, t) \in R^{n+1}, \|X\|_2 \leq t\}$
 $\Rightarrow$ SOCP

③ The semi-definite cone, $S_+^n = \{X | X = X^T \succeq 0\}$
 $\Rightarrow$ SDP

P_0 minimize $f_0(X)$
 subject to $f_i(X) \leq 0, AX=b$ mathematical form

$P_0 \xrightarrow{CVX} P_1$ SDPT3 Sedumi interior-point
 left. 1-3 SCS $\rightarrow$ ADMM $\checkmark$
 convex set, function, problem left. 4 solution
KKT condition

Semidefinite Programming (SDP)

$$\begin{array}{ll} \underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & x_1 F_1 + x_2 F_2 + \cdots + x_n F_n \preceq G \\ & Ax = b \end{array}$$

Handwritten notes in red: $A(x)$ and $G - A(x) \succeq 0$. A blue circle highlights the $\preceq$ symbol in the LMI constraint.

- Inequality constraint is called linear matrix inequality (LMI)
- Convex problem: linear objective and linear matrix inequality (LMI) constraints
- Observe that multiple LMI constraints can always be written as a single one

SDP I

• LP and equivalent SDP:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & \underline{Ax \preceq b} \end{array}$$

$$\begin{array}{ll} \underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & \underline{\text{diag}(Ax - b) \preceq 0} \end{array}$$

• SOCP and equivalent SDP:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f^T x \\ \text{subject to} & \underline{\|A_i x + b_i\| \leq c_i^T x + d_i, \quad i = 1, \dots, m} \end{array}$$

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f^T x \\ \text{subject to} & \underline{\begin{bmatrix} (c_i^T x + d_i)I & A_i x + b_i \\ A_i x + b_i & c_i^T x + d_i \end{bmatrix} \succeq 0, \quad i = 1, \dots, m} \end{array}$$

SDP II

• Eigenvalue minimization:

$$\underset{x}{\text{minimize}} \quad \lambda_{\max}(A(x))$$

where $A(x) = A_0 + x_1 A_1 + \cdots + x_n A_n$, is equivalent to SDP

$$\begin{array}{ll} \underset{x, t}{\text{minimize}} & t \\ \text{subject to} & A(x) \preceq tI \end{array}$$

• It follows from

$$\lambda_{\max}(A(x)) \leq t \iff A(x) \preceq tI$$

$$A(x) - tI \preceq 0$$

$$\lambda_i - t \leq 0, \forall i$$

Reference

Chapter 4 of:

- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University Press, 2004.

Lagrange Duality

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Outline

- 1 Lagrangian
- 2 Dual Function
- 3 Dual Problem
- 4 Weak and Strong Duality
- 5 KKT conditions

Lagrangian

- Consider an optimization problem in standard form (not necessarily convex)

$$\begin{array}{ll}\underset{\boldsymbol{x}}{\text{minimize}} & f_0(\boldsymbol{x}) \\ \text{subject to} & f_i(\boldsymbol{x}) \leq 0 \quad i = 1, \dots, m \\ & h_i(\boldsymbol{x}) = 0 \quad i = 1, \dots, p\end{array}$$

with variable $\boldsymbol{x} \in \mathbb{R}^n$, domain $\mathcal{D}$, and optimal value p^*

- The *Lagrangian* is a function $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$, with $\text{dom } L = \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p$, defined as

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\boldsymbol{x}) + \sum_{i=1}^m \lambda_i f_i(\boldsymbol{x}) + \sum_{i=1}^p \nu_i h_i(\boldsymbol{x})$$

where λ_i is the Lagrange multiplier associated with $f_i(\boldsymbol{x}) \leq 0$ and ν_i is the Lagrange multiplier associated with $h_i(\boldsymbol{x}) = 0$.

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- 1 Lagrangian
- 2 Dual Function**
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Lagrange Dual Function I

- The *Lagrange dual function* is defined as the infimum of the Lagrangian over $\mathbf{x}$: $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$,

$$\begin{aligned} g(\boldsymbol{\lambda}, \boldsymbol{\nu}) &= \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \\ &= \inf_{\mathbf{x} \in \mathcal{D}} \left(f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x}) \right) \end{aligned}$$

- Observe that:
 - the infimum is unconstrained (as opposed to the original constrained minimization problem)
 - g is concave regardless of original problem (infimum of affine functions)
 - g can be $-\infty$ for some $\boldsymbol{\lambda}, \boldsymbol{\nu}$

Lagrange Dual Function II

• **Lower bound property:** if $\lambda \succeq \mathbf{0}$, then $g(\lambda, \nu) \leq p^*$.

Proof.

Suppose $\tilde{x}$ is feasible and $\lambda \succeq \mathbf{0}$. Then,

$$f_0(\tilde{x}) \geq L(\tilde{x}, \lambda, \nu) \geq \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = g(\lambda, \nu)$$

Now choose minimizer of $f_0(\tilde{x})$ over all feasible $\tilde{x}$ to get $p^* \geq g(\lambda, \nu)$. $\square$

• We could try to find the best lower bound by maximizing $g(\lambda, \nu)$.
This is in fact the dual problem.

Outline

- 1 Lagrangian
- 2 Dual Function
- 3 Dual Problem**
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Dual Problem

- The *Lagrange dual problem* is defined as

$$\begin{array}{ll}\text{maximize} & g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \\ \text{subject to} & \boldsymbol{\lambda} \succeq \mathbf{0}\end{array}$$

- This problem finds the best lower bound on p^* obtained from the dual function
- It is a convex optimization (maximization of a concave function and linear constraints)
- The optimal value is denoted d^*
- $\boldsymbol{\lambda}, \boldsymbol{\nu}$ are dual feasible if $\boldsymbol{\lambda} \succeq \mathbf{0}$ and $(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \text{dom } g$ (the latter implicit constraints can be made explicit in problem formulation)

Example: Least-Norm Solution of Linear Equations I

• Consider the problem

$$\begin{array}{ll} \underset{x}{\text{minimize}} & x^T x \\ \text{subject to} & Ax = b \end{array}$$

• The Lagrangian is

$$L(x, \nu) = x^T x + \nu^T (Ax - b)$$

• To find the dual function, we need to solve an unconstrained minimization of the Lagrangian. We set the gradient equal to zero

$$\nabla_x L(x, \nu) = 2x + A^T \nu = 0 \implies x = -\frac{1}{2} A^T \nu$$

Example: Least-Norm Solution of Linear Equations II

and we plug the solution in L to obtain g :

$$g(\boldsymbol{\nu}) = L\left(-\frac{1}{2}\mathbf{A}^T\boldsymbol{\nu}, \boldsymbol{\nu}\right) = -\frac{1}{4}\boldsymbol{\nu}^T \mathbf{A}\mathbf{A}^T\boldsymbol{\nu} - \mathbf{b}^T\boldsymbol{\nu}$$

- The function g is, as expected, a concave function of $\boldsymbol{\nu}$.
- From the lower bound property, we have

$$p^* \geq -\frac{1}{4}\boldsymbol{\nu}^T \mathbf{A}\mathbf{A}^T\boldsymbol{\nu} - \mathbf{b}^T\boldsymbol{\nu} \text{ for all } \boldsymbol{\nu}$$

- The dual problem is the QP

$$\underset{\boldsymbol{\nu}}{\text{maximize}} \quad -\frac{1}{4}\boldsymbol{\nu}^T \mathbf{A}\mathbf{A}^T\boldsymbol{\nu} - \mathbf{b}^T\boldsymbol{\nu}$$

Example: Standard Form LP I

• Consider the problem

$$\begin{array}{ll}\underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0\end{array}$$

• The Lagrangian is

$$\begin{aligned}L(x, \lambda, \nu) &= c^T x + \nu^T (Ax - b) - \lambda^T x \\ &= (c + A^T \nu - \lambda)^T x - b^T \nu\end{aligned}$$

• L is a linear function of x and it is unbounded if the term multiplying x is nonzero.

Example: Standard Form LP II

• Hence, the dual function is

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \begin{cases} -\boldsymbol{b}^T \boldsymbol{\nu} & \boldsymbol{c} + \boldsymbol{A}^T \boldsymbol{\nu} - \boldsymbol{\lambda} = \mathbf{0} \\ -\infty & \text{otherwise} \end{cases}$$

• The function g is a concave function of $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ as it is linear on an affine domain.

• From the lower bound property, we have

$$p^* \geq -\boldsymbol{b}^T \boldsymbol{\nu} \quad \text{if } \boldsymbol{c} + \boldsymbol{A}^T \boldsymbol{\nu} \succeq \mathbf{0}$$

• The dual problem is the LP

$$\begin{array}{ll} \underset{\boldsymbol{\nu}}{\text{maximize}} & -\boldsymbol{b}^T \boldsymbol{\nu} \\ \text{subject to} & \boldsymbol{c} + \boldsymbol{A}^T \boldsymbol{\nu} \succeq \mathbf{0} \end{array}$$

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- 1 Lagrangian
- 2 Dual Function
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- 4 Weak and Strong Duality**
- 5 KKT conditions

Weak and Strong Duality I

- From the lower bound property, we know that $g(\lambda, \nu) \leq p^*$ for feasible (λ, ν) . In particular, for a (λ, ν) that solves the dual problem.
- Hence, **weak duality** always holds (even for nonconvex problems):

$$d^* \leq p^*$$

- The difference $p^* - d^*$ is called **duality gap**.
- Solving the dual problem may be used to find nontrivial lower bounds for difficult problems.
- Even more interesting is when equality is achieved in weak duality. This is called **strong duality**:

$$d^* = p^*$$

Weak and Strong Duality II

- Strong duality means that the duality gap is zero.
- Strong duality:
 - is very desirable (we can solve a difficult problem by solving the dual)
 - does not hold in general
 - usually holds for convex problems
 - conditions that guarantee strong duality in convex problems are called **constraint qualifications**.

Slater's Constraint Qualification I

- Slater's constraint qualification is a very simple condition that is satisfied in most cases and ensures strong duality for convex problems.
- Strong duality holds for a convex problem

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & f_0(\mathbf{x}) \\ \text{subject to} & f_i(\mathbf{x}) \leq 0 \quad i = 1, \dots, m \\ & \mathbf{Ax} = \mathbf{b} \end{array}$$

if it is strictly feasible, i.e.,

$$\exists \mathbf{x} \in \text{int } \mathcal{D} : \quad f_i(\mathbf{x}) < 0 \quad i = 1, \dots, m, \quad \mathbf{Ax} = \mathbf{b}$$

- There exist many other types of constraint qualifications.

Example: Inequality Form LP

- Consider the problem

$$\begin{array}{ll}\underset{x}{\text{minimize}} & c^T x \\ \text{subject to} & Ax \preceq b\end{array}$$

- The dual problem is

$$\begin{array}{ll}\underset{\lambda}{\text{maximize}} & -b^T \lambda \\ \text{subject to} & A^T \lambda + c = 0, \quad \lambda \succeq 0\end{array}$$

- From Slater's condition: $p^* = d^*$ if $A\tilde{x} \prec b$ for some $\tilde{x}$.
- In this case, in fact, $p^* = d^*$ except when primal and dual are infeasible.

Example: Convex QP

- Consider the problem (assume $P \succeq 0$)

$$\begin{array}{ll}\underset{x}{\text{minimize}} & x^T P x \\ \text{subject to} & Ax \preceq b\end{array}$$

- The dual problem is

$$\begin{array}{ll}\underset{\lambda}{\text{maximize}} & -\frac{1}{4}\lambda^T A P^{-1} A^T \lambda - b^T \lambda \\ \text{subject to} & \lambda \succeq 0\end{array}$$

- From Slater's condition: $p^* = d^*$ if $A\tilde{x} \prec b$ for some $\tilde{x}$.
- In this case, in fact, $p^* = d^*$ always.

Complementary Slackness

- Assume strong duality holds, $\mathbf{x}^*$ is primal optimal and $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is dual optimal. Then

$$\begin{aligned} f_0(\mathbf{x}^*) = g(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) &= \inf_{\mathbf{x}} \left(f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i^* f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i^* h_i(\mathbf{x}) \right) \\ &\leq f_0(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* f_i(\mathbf{x}^*) + \sum_{i=1}^p \nu_i^* h_i(\mathbf{x}^*) \\ &\leq f_0(\mathbf{x}^*) \end{aligned}$$

- Hence, the two inequalities must hold with equality. Implications:
 - $\mathbf{x}^*$ minimizes $L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$
 - $\lambda_i^* f_i(\mathbf{x}^*) = 0$ for $i = 1, \dots, m$; this is called **complementary slackness**:

$$\lambda_i^* > 0 \implies f_i(\mathbf{x}^*) = 0, \quad f_i(\mathbf{x}^*) < 0 \implies \lambda_i^* = 0$$

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Karush-Kuhn-Tucker (KKT) Conditions

KKT conditions (for differentiable f_i, h_i):

1 primal feasibility:

$$f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \quad h_i(\mathbf{x}) = 0, \quad i = 1, \dots, p$$

2 dual feasibility: $\lambda \succeq \mathbf{0}$

3 complementary slackness: $\lambda_i f_i(\mathbf{x}) = 0$ for $i = 1, \dots, m$

4 zero gradient of Lagrangian with respect to $\mathbf{x}$:

$$\nabla f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i \nabla f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i \nabla h_i(\mathbf{x}) = \mathbf{0}$$

KKT condition

- We already know that if strong duality holds and x, λ, ν are optimal, then they must satisfy the KKT conditions.
- What about the opposite statement?
- If x, λ, ν satisfy the KKT conditions for a convex problem, then they are optimal.

Proof.

From complementary slackness, $f_0(x) = L(x, \lambda, \nu)$ and, from 4th KKT condition and convexity, $g(\lambda, \nu) = L(x, \lambda, \nu)$. Hence, $f_0(x) = g(\lambda, \nu)$. $\square$

Theorem

If a problem is convex and Slater's condition is satisfied, then x is optimal if and only if there exists λ, ν that satisfy the KKT conditions.

Reference

Chapter 5 of:

- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University Press, 2004.